Integrated Personnel and Pay System – Army (IPPS-A)

IPPS-A Provisioning Plan (PP)

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Table of Contents

1 Introduction and Purpose 1
1.1 Introduction 1
1.2 Document Purpose 1
1.3 Document Overview 1
1.4 Reference Documents 1

2 Identity Management Architecture Core IPPS-A 2
2.1 PeopleTools 2
2.2 Oracle Unified Directory Load Application Engine 2

3 Identity and Access Management 4
3.1 Managing Identities 5
3.2 Authentication 5
3.2.1 Common Access Card 5
3.2.2 User ID and Password 5
3.2.3 Single Sign-On 5
3.2.4 User Access Control 6
3.2.5 Categories/Subcategories 6

4 Access and Role Management 7
4.1 Attribute Based Access Control 7
4.2 Roles Based Access Control 8

5 Provisioning Management 10
5.1 Target Applications 11
5.2 Provisioning Types 11
5.3 User Access Request 11
5.4 Recertification 13
5.5 Provisioning Functionality 13
5.6 Provisioning System Components 14
5.7 Provisioned User Types 14
5.7.1 Member 14
5.7.2 Person of Interest 14
5.8 Provisioned Access Level 15
5.8.1 Self-Service User 15
5.8.2 Authorized User 15
5.8.3 Privileged User 15
5.9 Reconciliation 15
5.10 Trusted Sources 16

6 Managing User Request Process and Workflow Approval 17
Request Process Components 17

7 Audit Management 18
7.1 Segregation of Duty 18

8 Testing Strategy 19
8.1 Test Script/Scenario Development Standard 19
8.2 Provision Testing 19

9 Deployment Strategy 20

Appendix A: Acronyms and Abbreviations A-1

Appendix B: CACI Consolidated Reference Documents B-1

Appendix C: Glossary C-1

List of Figures

Figure 2-1 PeopleSoft User Profile Sync.....2

Figure 3-1 High-Level Authentication/Authorization Process 4

Figure 3-2 Breakdown of the Seven End User Categories into IPPS-A Actions 6

Figure 4-1 ABAC Methodology 7

Figure 4-2 RBAC Methodology 9

Figure 5-1 Automated Provisioning New Army Member into IPPS-A.....10

Figure 5-2 User Access Request Workflow.....12

Figure 5-3 User Access Request Workflow for National Guard 13

List of Tables

Table 1-1 Reference Documentation 1

Table 3-1 Description of High-Level Authentication/Authorization Process Elements 4

Table 8-1 Test Phases 19

1 INTRODUCTION AND PURPOSE

1.1 Introduction

The Army has developed and implemented the Integrated Personnel and Pay System – Army (IPPS-A), a Web-based tool, available 24 hours a day, accessible to Soldiers, Human Resources (HR) professionals, Combatant Commanders, personnel and pay managers, and other authorized users throughout the Army.

Team CACI has been selected as the System Integrator (SI) to maintain successful deployment of IPPS-A.

1.2 Document Purpose

This document describes the approach for provisioning users into IPPS-A. The Provisioning Process will use Application Security and identity management components to provision user profiles with the appropriate access attributes (roles and permissions) across the IPPS-A applications: PeopleSoft PeopleTools, Human Capital Management (HCM), Customer Relationship Management (CRM), Enterprise Learning Management (ELM), and Oracle Business Intelligence Enterprise Edition (OBIEE).

1.3 Document Overview

This document applies to all activities for the IPPS-A Increment II, Provisioning approach in NIPR and SIPR. This document addresses PeopleSoft PeopleTools, Oracle Identity Access Management (OIAM) architecture components, provisioning management, Role Based Access Control (RBAC), Attribute Based Access Control (ABAC), and the user access request process flow.

1.4 Reference Documents

Table 1-1 lists key IPPS-A reference documents.

Table 1-1 Reference Documentation

Document	Date
Application Security and Development Security Technical Implementation Guide (STIG), Version 6, Release 2	28 JAN 25
Army Regulation (AR) 25-2	04 APR 19
Fusion Middleware Integration Guide for Oracle Identity Management Suite 12c (12.2.1.4.0)	APR 23
IPPS-A Contract W91CRB-25-C-5007, Mod 001	31 DEC 24
IPPS-A Increment II Requirements Glossary and Verb List, Version 1.5	20 AUG 19
Oracle Identity Manager - Connector Guide for PeopleSoft User Management (E25371-18)	APR 22
PeopleSoft PeopleTools 8.60	OCT 22

2 IDENTITY MANAGEMENT ARCHITECTURE CORE IPPS-A

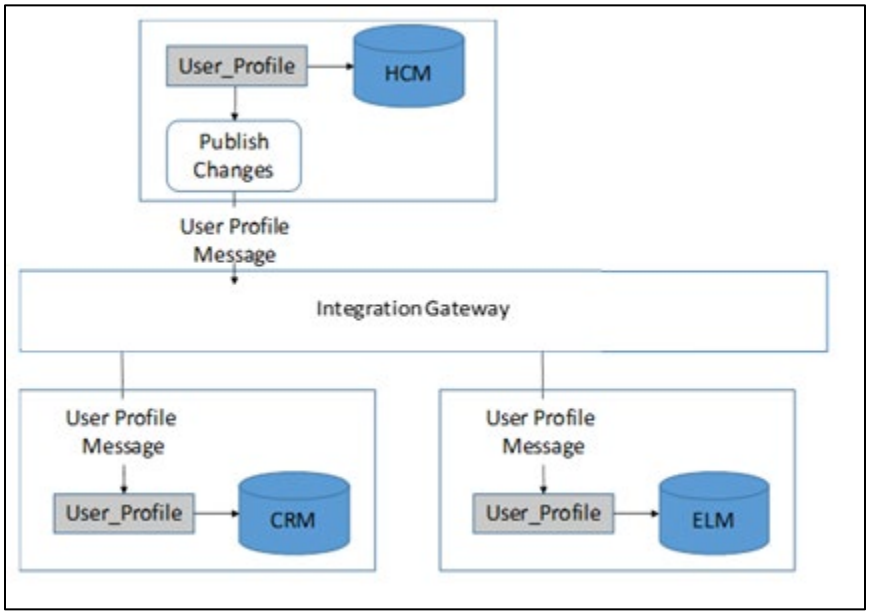
IPPS-A will use PeopleTools and the Oracle Unified Directory Load application engine as core components to implement provisioning for all users that require access. With IPPS-A's unique and complex functional requirements, PeopleTools and the IP_OUD_LOAD app engine will integrate with infrastructure, application, and business processes.

2.1 PeopleTools

PeopleTools is the development and runtime platform for PeopleSoft applications. Its features include standards-based integration, robust reporting, advanced user interface features, administrative tools, and lifecycle management tools. PeopleTools facilitates the deployment of both vendor-developed and custom-developed applications using an Internet-based architecture known as the PeopleSoft Internet Architecture (PIA). PIA consists of a variety of components, including web, application, search, database, and process scheduler servers as well as Integration Broker.

Leveraging delivered integration features such as user profile synchronization provides a swift method of identity management provisioning for customers of multiple PeopleSoft environments where the user logins are the same. User profile synchronization involves setting up each PeopleSoft database in the enterprise to send and receive user profile updates through the Integration Broker. Figure 2-1 shows the service operations and messages, and the way in which user profile data is published by HCM and subscribed to by CRM, and ELM PeopleSoft systems that use default user profile synchronization.

Figure 2-1 PeopleSoft User Profile Sync



2.2 Oracle Unified Directory Load Application Engine

This application engine runs every 12 hours and has replaced Oracle Identity Management. It updates Oracle Unified Directory (OUD) based on PeopleSoft HCM data. This application engine will reconcile with the additional PeopleSoft applications such as ELM and CRM. Further

details on the IP_OUD_LOAD app engine will be covered in CDRL 03-006 IPPS-A Provisioning Design.

3 IDENTITY AND ACCESS MANAGEMENT

The Identity and Access Management System identifies and centrally manages data in IPPS-A to authenticate users and grant access rights to system resources and data based on user roles and groups. IPPS-A implements Oracle Access Manager (OAM) and the provisioning management process to provide each Member and Person of Interest (POI) with the appropriate access.

Figure 3-1 describes the authentication/authorization process for IPPS-A users.

Figure 3-1 High-Level Authentication/Authorization Process

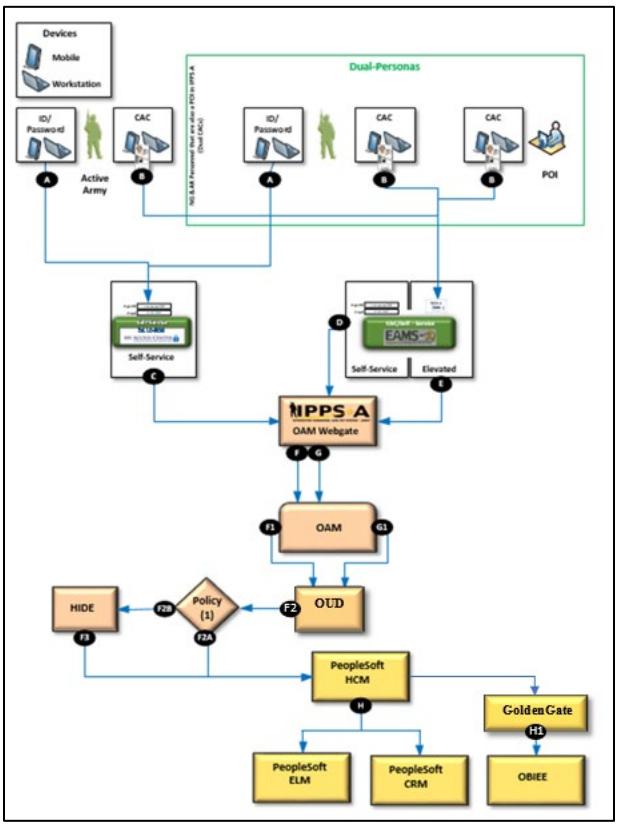


Table 3-1 describes the elements of the High-Level Authentication/Authorization Process.

Table 3-1 Description of High-Level Authentication/Authorization Process Elements

Label/ Object	Actor	Description
A:	Active Army/NG/Reserve User:	Uses their Mobile device to login to DS Logon with ID/Password.
B:	Active Army/NG/Reserve/POI User:	Uses their Mobile Device or Workstation to login with their CAC.
B:	Active Army/NG/Reserve User:	Uses their Workstation to login to EAMS-A with their CAC.
C:	Active Army/NG/Reserve User:	Clicks IPPS-A Link on DS Logon Portal Page.
D:	Active Army/NG/Reserve User:	Clicks IPPS-A link on EAMS-A Portal Page.
E:	Active Army/NG/Reserve/POI User:	Clicks IPPS-A link on EAMS-A Portal Page.
F:	Active Army/NG/Reserve User:	Authentication with ID/Password.
F1:	Active Army/NG/Reserve User:	Authentication with ID/Password detected.
F2:	Active Army/NG/Reserve User:	Access Authorization.

Label/ Object	Actor	Description
Policy (1):		Policy evaluates if the user is currently a privileged user in IPPS-A.
F2A:		If Policy (1) evaluation is false, the process passes the user through to the IPPS-A Portal.
F2B:		If Policy (1) evaluation is true, hide privileged navigation from the user.
F3:	Active Army/NG/Reserve User:	Only access Self-Service authorization. [Access: Self-Service only]
G:	Active Army/NG/Reserve/POI User:	Authentication with CAC.
G1:	Active Army/NG/Reserve/POI User:	Authentication with CAC detected.
G2:	Active Army/NG/Reserve User:	Access Authorization with elevated privileges.
H:	User [Access to IPPS-A Applications]	
H1:	User [Access to IPPS-A Applications]	Reconciliation process from GoldenGate to OBIEE.

3.1 Managing Identities

IPPS-A integrates with Army-Approved Enterprise Services to manage digital identities, the network, or Internet equivalent to the real identity of a Member/POI for identification and authentication connections. The primary purpose of managing identities is to bind a digital identity to that user uniquely, which is used across the IPPS-A suite.

3.2 Authentication

Authentication verifies the digital identity of a user. In this process, the user submits credentials and Army-Approved Enterprise Services validates the credentials as proof of identity. IPPS-A Increment II accepts a Common Access Card (CAC) and personal identification (ID) number (PIN) or user ID and password as valid user credentials with defined access controls applied to each user according to the type of credentials used for login.

3.2.1 Common Access Card

A CAC and PIN are required when accessing IPPS-A. IPPS-A supports CAC and PIN authentication by leveraging these credentials from the Army-Approved Enterprise Service.

3.2.2 User ID and Password

User ID and password are only used when accessing IPPS-A to perform self-service user functions. IPPS-A supports user ID and password authentication by using these credentials from the Department of Defense (DoD) Self-Service Logon (DS Logon).

3.2.3 Single Sign-On

Single Sign-On (SSO) controls access to multiple applications with a single logon. By using this method, the user logs in once and gains access to all systems without being prompted to log in separately to other applications. In IPPS-A, after initially logging in to the Army-Approved Enterprise Service, SSO allows users to seamlessly transition into the IPPS-A PeopleSoft HCM and then to any other IPPS-A applications to which the user has access.

3.2.4 User Access Control

User access control is the capability to control user access to IPPS-A data from unauthorized access. Access is determined by establishing user accounts/user profiles accompanied by roles, permissions, and access control attributes.

Further information on these topics is in Contract Data Requirements List (CDRL) 03-007, IPPS-A Roles and Permissions Guide, and CDRL 03-004, IPPS-A Application Security Design.

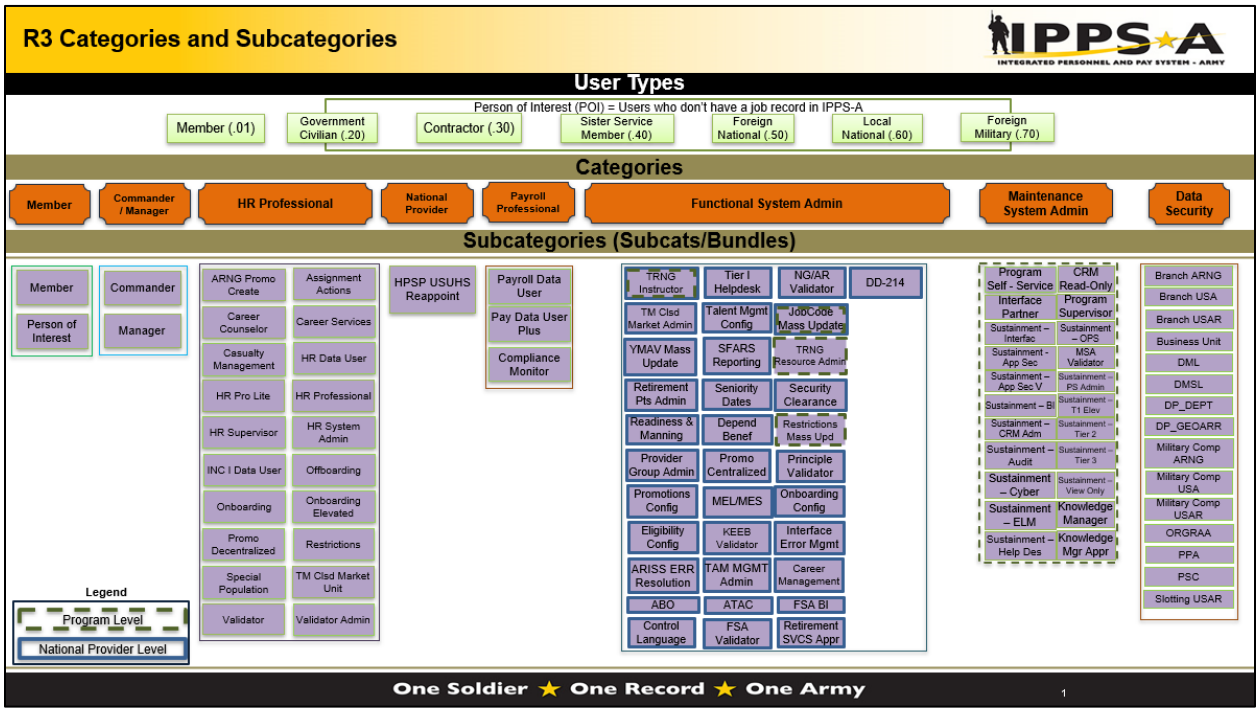
3.2.5 Categories/Subcategories

Categories are groupings of subcategories. A subcategory is a logical collection of user functions that share one or more common attributes. For IPPS-A, implementing subcategories simplifies the selection of PeopleSoft roles required to perform a job. Roles are grouped into subcategories to manage users and roles efficiently.

Figure 3-2 shows all current Categories and Subcategories for security access.

Further information on User Profile Group is in CDRL 03-006, IPPS-A Provisioning Design.

Figure 3-2 Breakdown of the Seven End User Categories into IPPS-A Actions



4 Access and Role Management

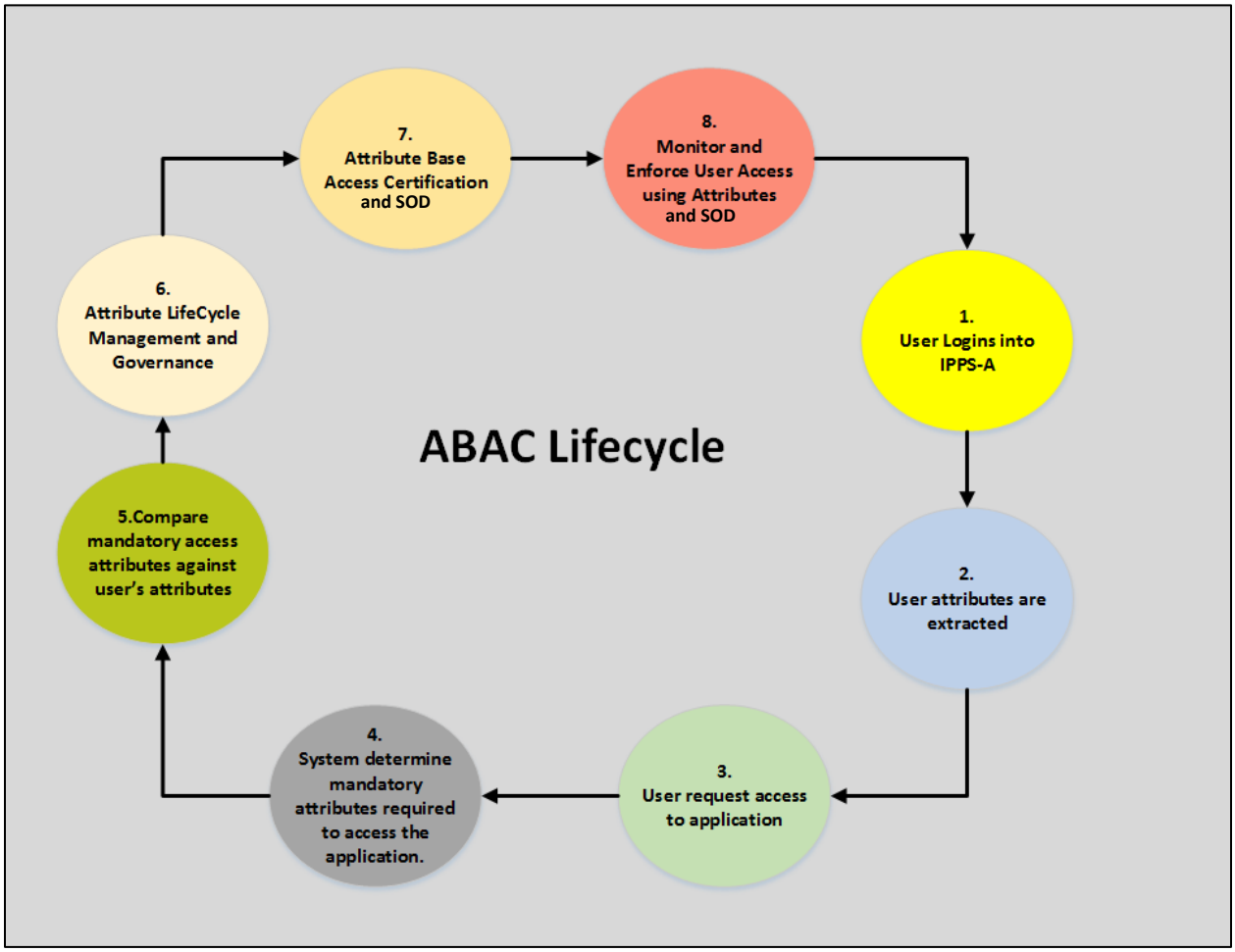
IPPS-A leverages the methodologies of both ABAC and RBAC to ensure that roles and security attributes protect sensitive data. Figure 4-1 describes the ABAC methodology.

4.1 Attribute Based Access Control

ABAC is a logical access-control model distinguished by controlling access to objects based on the evaluation of business rules against the attributes of the entities (subject and objects). ABAC uses attributes as building blocks ultimately to construct access request templates as a business user-case policy. ABAC attributes are sets of properties that can be used to determine all the entities that must be considered for authorization purposes.

ABAC uses three different attribute types: User Attributes (employee type, UIC, training course), Application Attributes (HCM, ELM, CRM), and System Attributes (time zone, sign-on time, timeout). For example, IPPS-A users can be granted only specified HR roles after passing required ELM training courses, which update the attribute of the training course to a status of "complete." ABAC will be used in IPPS-A in conjunction with business rules to assign roles and permissions to entities.

Figure 4-1 ABAC Methodology



The ABAC Lifecycle Methodology sequence is:

1. User logins into IPPS-A.
2. The system extracts user attributes from the database.
3. The user requests access to an application with specific user type profile.
4. The system determines mandatory attributes required to access the application.
5. The system compares mandatory attributes against the user's attributes.
6. As part of role governance, review and update attributes as required.
7. Achieve compliance by way of access certifications and segregation of duty (SOD).
8. Monitor and enforce user access using attributes and SOD.

4.2 Roles Based Access Control

IPPS-A Access and Role Management leverages RBAC methodology, a method of assigning and regulating access to application resources based on the roles of individual users across the IPPS-A enterprise. For example, IPPS-A grants a user access to PeopleSoft HCM functionality based on the PeopleSoft HCM Role assigned to a user. The PeopleSoft HCM Role provides a user with access only to functionality (components, menus, pages, etc.) that the user needs to complete the normal duties of a user's job according to the Least Privilege principle.

The concept of limiting access, or "least privilege," is to provide authorization to perform required functions only. The goal of Least Privilege is to reduce risk by limiting the number of individuals with access to critical system security controls, controlling who is allowed to enable or disable system security features or change privileges of users or programs.

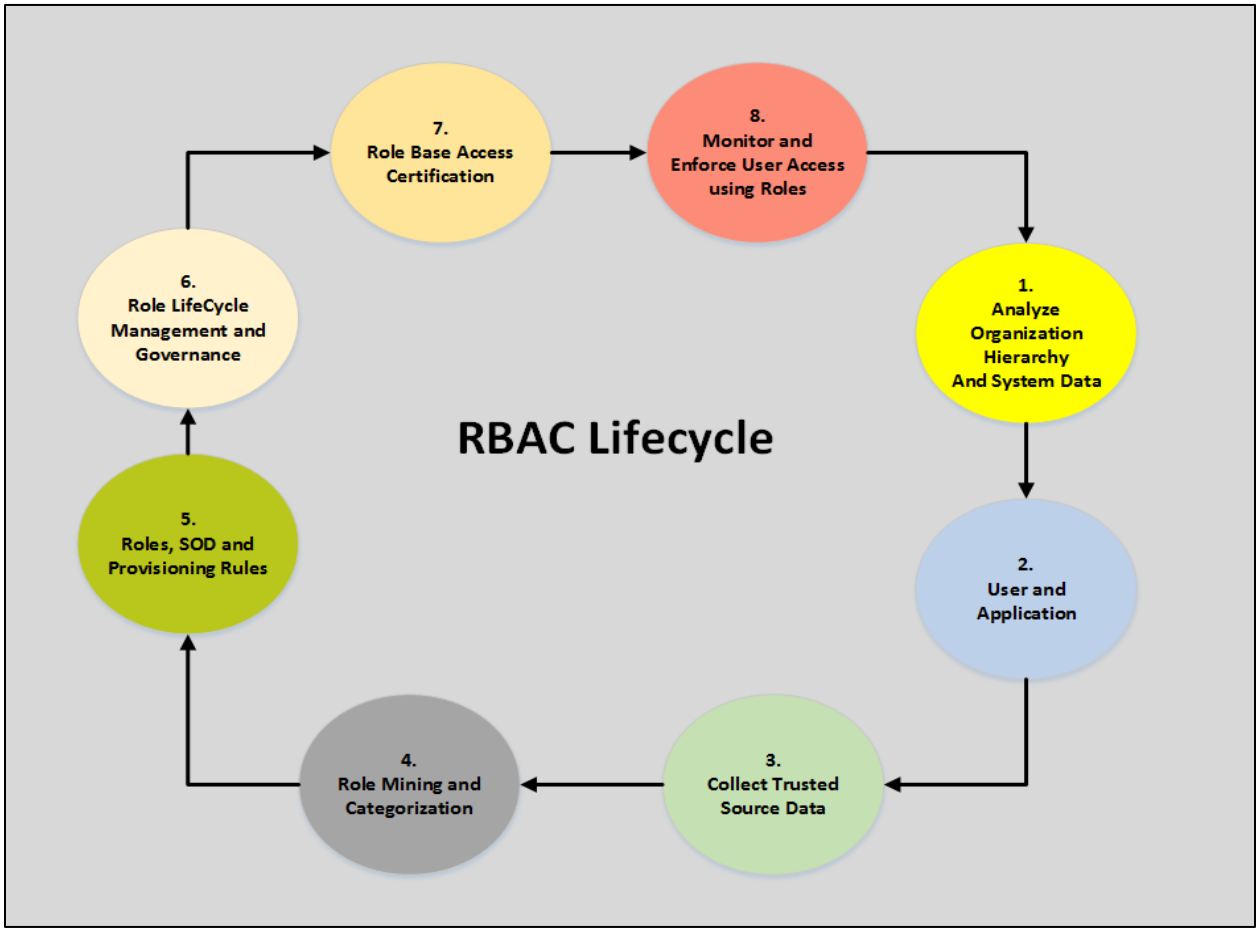
Role definition is a critical step in deploying any RBAC system. Roles can be defined at an abstract level from a business perspective (Top-down), or context-specific to an application or system from a technology perspective (Bottom-up). Both Top-down and Bottom-up approaches have distinct challenges. For instance, the Top-down approach does not provide easy enforcement capabilities. Understanding roles may not be possible without understanding the entire context. Conversely, the Bottom-up approach lacks the perspective of, or input from, business owners.

The IPPS-A solution to role definitions is a hybrid between the Bottom-up and Top-down approaches. This hybrid approach leverages core roles from Bottom-up role mining, aligning them to job functions from Top-down analysis.

IPPS-A's implementation will derive user resource access via a RBAC methodology that limits access to system applications to only authorized users in an organization. For IPPS-A, implementing an RBAC methodology simplifies identity and access control compliance. The RBAC methodology manages access based on a user's functional job tasks within the hierarchal structure of IPPS-A's organization, not on an individual, user-by-user basis.

Figure 4-2 describes the RBAC Methodology:

Figure 4-2 RBAC Methodology

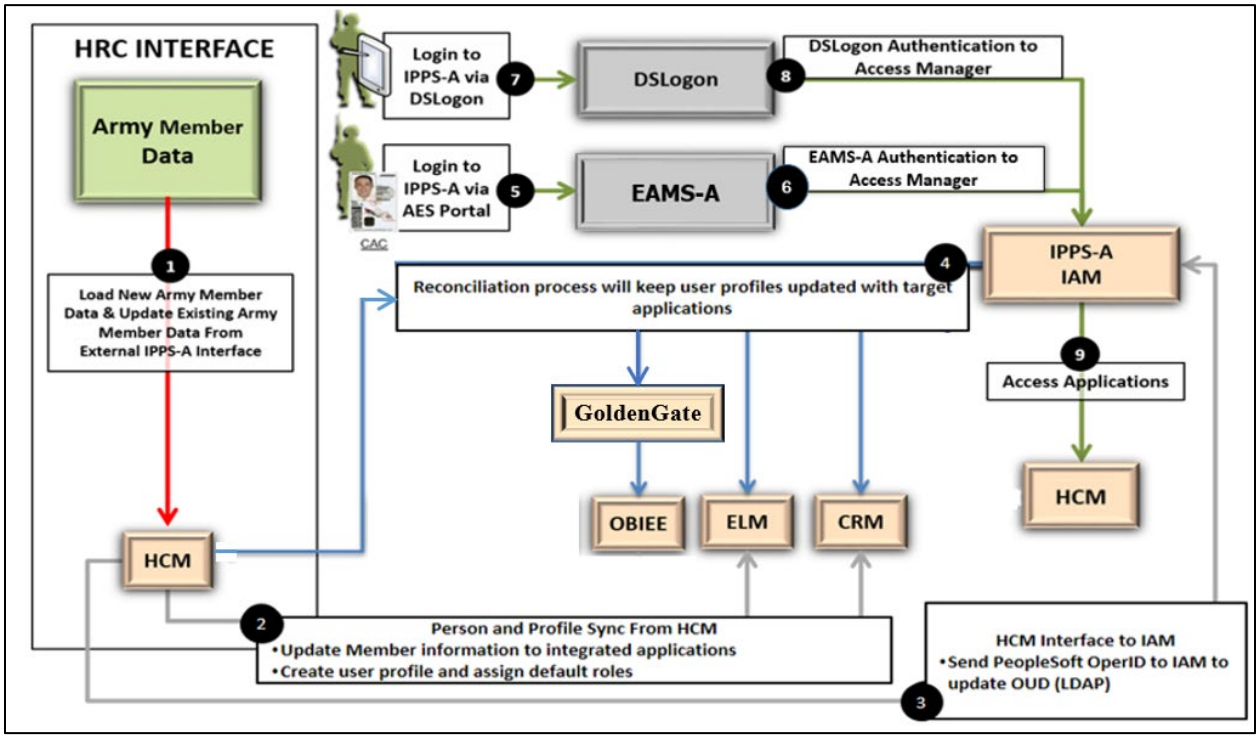


1. The RBAC Lifecycle Methodology sequence is:
2. Analyze organization hierarchy system data.
3. Gather user and application data.
4. Collect data from trusted source.
5. Analyze roles and categorization.
6. As part of role definition, review and finalize roles. Review and define Segregation of Duties (SOD) rules for provisioning.
7. As part of role governance, review and update roles as required.
8. Achieve compliance by way of access certifications and SOD.
9. Monitor and enforce user access using roles and SOD.

5 PROVISIONING MANAGEMENT

IPPS-A defines provisioning as the process and capability to create, modify, or deactivate user accounts and roles in a target system. Provisioning depends on having some endpoint destination. The endpoint in a provisioning environment is known as the target or "target/targeted system." Targets are software applications that serve as the endpoint destination. IPPS-A will initiate these processes via a self-service request or automated provisioning process that ends at the target systems, PeopleSoft HCM, CRM and ELM. IPPS-A provisioning is an RBAC implementation; therefore, user functionality PeopleSoft is determined by the user's Membership based on roles assigned to the user. Access to OBIEE will be via verification with GoldenGate. Figure 5-1 provides the automated provisioning process of a new Army Member into IPPS-A.

Figure 5-1 Automated Provisioning New Army Member into IPPS-A



IPPS-A Automated Process to Provision an Army Member:

1. This step will load new Army Member data and update existing Army Member data from external IPPS-A interface.
2. If new Army Member, create user profile and assign default roles. If existing Army Member, update user's profile attributes.
   a. Person and Profile Sync interfaces are utilized to provision to ELM and CRM.
3. This step is used for the synchronization (sync) process to push user identity automatically with the IP_OUD_LOAD app engine and then to update OUD.
4. This step is used for the synchronization (sync) process to push user roles from HCM to a separate copy of HCM that GoldenGate creates, which then updates OBIEE.

5. In this step, the Member logs into IPPS-A via the Army Enterprise Service (AES).
6. In this step, Identity Access Management (IAM) authenticates with Enterprise Access Management Service - Army (EAMS-A).
7. In this step, the Member logs into IPPS-A via DoD Self-Service Logon (DS Logon).
8. In this step, Identity Access Management (IAM) authenticates with DS Logon.
9. In this step, access to the applications is successful.

5.1 Target Applications

HCM populates target systems with data and provisions user accounts/user profiles and applicable information. The IP_OUD_LOAD app engine will reconcile all applications to ensure any changes to user profiles are kept in sync. The following are the target systems:

- PeopleSoft Human Capital Management (HCM)
- PeopleSoft Customer Relationship Management (CRM)
- PeopleSoft Enterprise Learning Management (ELM)

5.2 Provisioning Types

IPPS-A will implement the following two types of provisioning:

- Request-based provisioning: A request can be manually created by the administrator or by users themselves to provision the account in the target system. Approval workflows are started after a provisioning request is submitted, and after approval, the account is provisioned in the target system. A request-based provisioning operation involves both users and approvers. Typically, approvers are in the management chain of the requesters. Example operations include creating a user, provisioning an account, and granting a role to a user.
- Policy-based provisioning: In policy-based provisioning, the user is provisioned in the target resource automatically, based on defined rules, policies, and triggers. These policies can be driven by user roles, attributes, or conditions. By default, each Member of these roles receives a predefined account in the target application. Administrators can specify access levels for each resource provisioned, granting each user only the exact level of access required to complete the job.

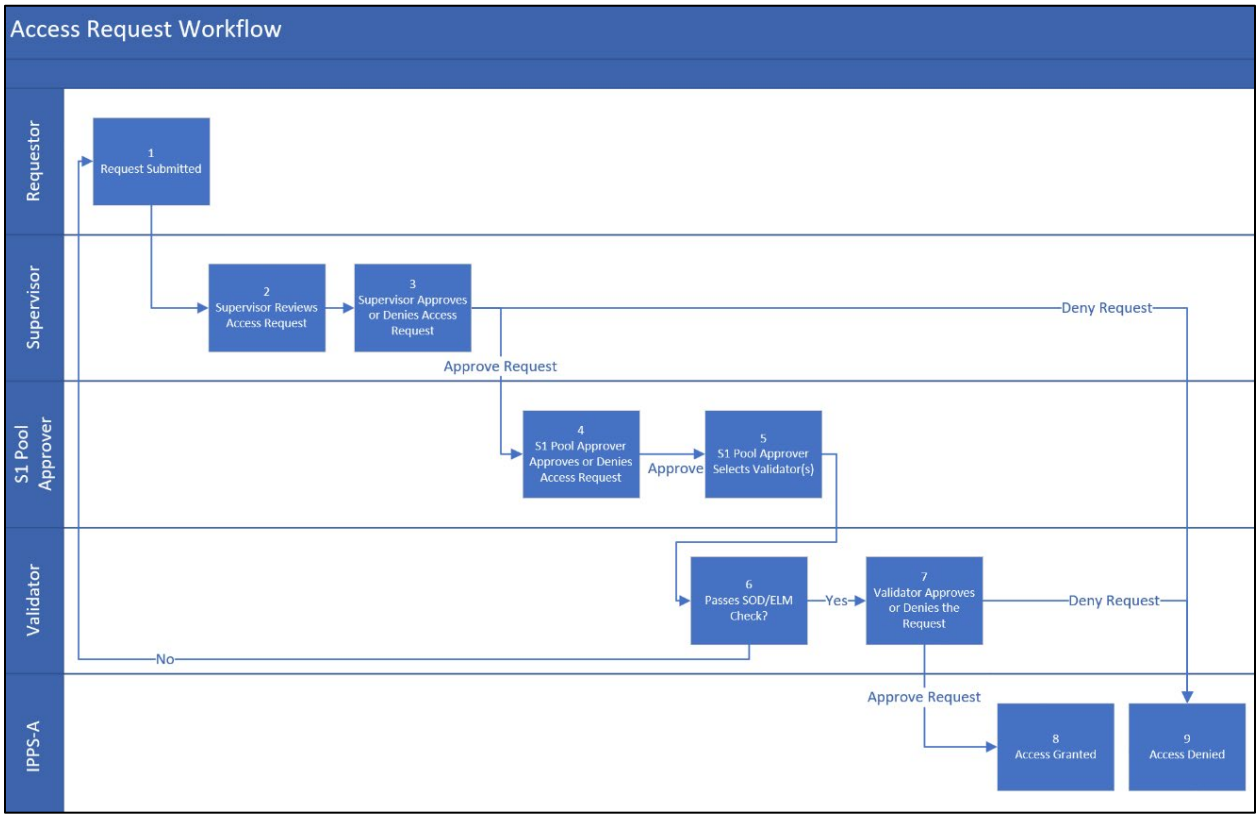
5.3 User Access Request

A User Access Request is an event that users or administrators create to perform a specific action that is required for users to perform their daily tasks. Various operations, such as requesting/adding a role or being provisioned to an application instance, is performed through user requests. User access requests for all IPPS-A user types are approved through a workflow process. The workflow process in Figure 5-2 graphically represents the self-service request a user (Member or POI) follows to gain access to an application.

A user access request can be initiated by a user or initiated on the user's behalf and starts by a requestor generating an online IPPS-A Access Request, to request access formally to a resource (application, role, etc.). The user access request is a provisioning feature that enables IPPS-A to automate business processes that manage user access. The workflow process will route to

validators for review and assessment as well as validator administrators if/when SOD or training issues are found with the requestor.

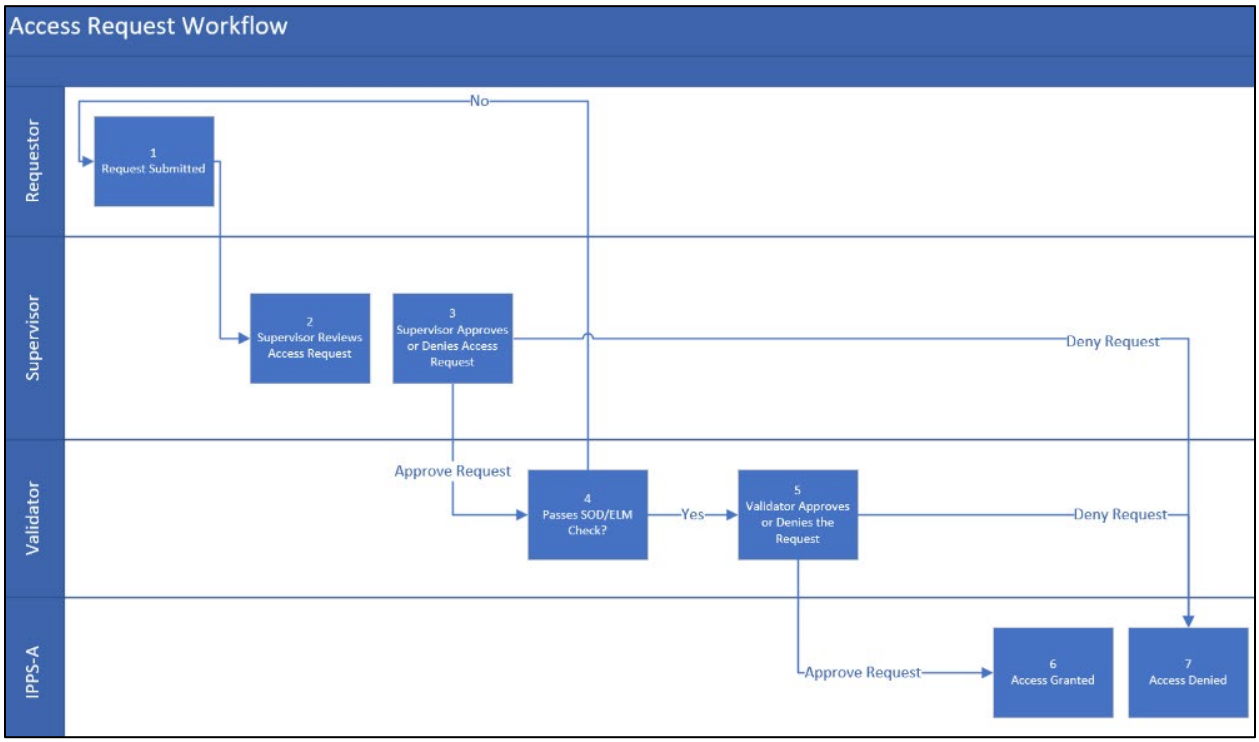
Figure 5-2 User Access Request Workflow



The IPPS-A Access Request Process is:

1. An initiator/requestor requests elevated privilege.
2. The user's Supervisor will review the access request.
3. The Supervisor will approve or deny the access request.
4. The S1 Pool Approver will approve or deny the access request.
5. If approved, the S1 Pool Approver selects a Validator(s) for approval. (Note: Further detail can be found in 03-006 Provisioning Design)
6. The Validator performs the SOD/ELM verification.
7. If there are no SOD conflicts or ELM issues, the Validator will action the access request.
8. If the Validator does not approve the request, it is rejected.
9. If the Validator approves the request, the user is granted the requested access.

Figure 5-3 User Access Request Workflow for National Guard



The IPPS-A Access Request Process for the National Guard is (Note - National Guard does not have S1 Pool Approvers):

1. An initiator/requestor requests elevated privilege.
2. The user's Supervisor will review the access request.
3. The Supervisor will approve or deny the access request.
4. The Validator performs the SOD/ELM verification.
5. If there are no SOD conflicts or ELM issues, the Validator will action the access request.
6. If the Validator does not approve the request, it is rejected.
7. If the Validator approves the request, the user is granted the requested access.

5.4 Recertification

When a user's access request is approved by a Validator, an access expiration date is set to one year by default. A Validator is able to select an expiration date less than one year if needed. Starting at 60 days before the access expiration date, the user will start receiving email notifications that they must submit a recertification request. Further details on the IPPS-A recertification process will be covered in CDRL 03-006 IPPS-A Provisioning Design.

5.5 Provisioning Functionality

Main functionality provided by the provisioning system is:

- Provisioning: Process of creating an account in the target system.
- De-provisioning: Process of deactivating a user account in the target system.

- Re-provisioning: Process of reactivating a user account for access to the target system.
- Reporting and auditing: The creation of documentation for the provisioning processes and enforcement. The documentation is essential for auditing, regulatory, and compliance purposes.

5.6 Provisioning System Components

The IPPS-A provisioning system uses the following PeopleSoft and OAM components:

- OAM: Manages user access to system resources.
- IP_OUD_LOAD Application Engine: Updates OUD based on data in PeopleSoft HCM.
- Identity Provisioning Services: Provision Workflow, Resource and Account Management, and Auto Group Membership.
- Integration Services: PeopleSoft Connector, Schedule Jobs, Identity Connector Framework (ICF), Adapter Factory, and Remote Manager.
- SOA: Workflow validator and notification service that provides an interactive environment to request, approve, and manage user access.
- WebLogic: The software application that used as a foundation and container environment used for managed servers (OAM and SOA).

OAM represents the IAM solution. IPPS-A's Provisioning System uses HCM to populate data in target systems from trusted sources (such as the HR Command [HRC] Interface) for reliable user data. The IP_OUD_LOAD app engine will be used to reconcile all PeopleSoft instances. GoldenGate will be used to reconcile OBIEE.

5.7 Provisioned User Types

The IPPS-A provisioning system provisions two types of users in the system: Members and POIs. The following subsections describe each type.

5.7.1 Member

For provisioning, a Member user type is an Army Soldier. By default, the Member user type is assigned the self-service role.

5.7.2 Person of Interest

For provisioning, a POI user type is similar to the Member provisioning process:

- POI is a user in IPPS-A in one of six categories:
  - Government Civilians (Federal and State): EMPLID.20
  - Contractors: EMPLID.30
  - U.S. Military (non-Army): EMPLID.40
  - Foreign Nationals: EMPLID.50
  - Local Nationals: EMPLID.60
  - Foreign Military: EMPLID.70
- POI will be an authorized user in the system.

- POI will be assigned POI self-service roles.
- POI requesting process to gain access to IPPS-A.
- POI has a record in IPPS-A to identify business unit and department association.

5.8 Provisioned Access Level

The provisioning process is based on the user type and level of access the user requires. In IPPS-A's provisioning, user-access types fit in three categories:

- Self-Service
- Authorized
- Privileged

5.8.1 Self-Service User

An IPPS-A Self-Service user has assigned roles and permissions to initiate requests or transactions that impact only his/her own record.

PeopleSoft user synchronization will automate privileges and provisioning of resources to various target systems for Self-Service users. The detailed process is in CDRL 03-006, IPPS-A Provisioning Design.

5.8.2 Authorized User

Authorized users can be internal or external individuals provisioned with roles that authorize them to execute more than self-service actions. For example, an HR professional will support transactions on members of his/her organization.

5.8.3 Privileged User

In IPPS-A, a Privileged user is an administrative user with elevated roles and administrative permissions assigned to that user type. In addition, an IPPS-A privileged/administrative user also has appropriate roles and permissions to perform self-service, security, and administrative functions and tasks.

5.9 Reconciliation

Reconciliation is the process by which operations—such as user creation, modification, or deactivation that start in the IP_OUD_LOAD app engine and communicates with the targeted system. The reconciliation process compares entries in the app engine and target system repository, identifies the differences between the two, and applies the latest changes made on the target system to the app engine. The reconciliation process within IPPS-A is used to populate and synchronize the data between the app engine and the targeted system. Further details on IPPS-A system account reconciliation will be covered in CDRL 03-006 IPPS-A Provisioning Design.

The IP_OUD_LOAD app engine implements the following two types of reconciliation:

- Trusted source reconciliation: If data is reconciled from a system (for example, OID, Oracle Unified Directory [OUD], or Active Directory) that initiates the creation of users, roles, role memberships that reconciliation mode is trusted source reconciliation, also known as authoritative source reconciliation.

- Account reconciliation: In account reconciliation mode, changes made to a user's access rights on an external resource are reconciled directly into OUD.

5.10 Trusted Sources

A trusted source is a system that has authoritative data that other systems can safely use. IPPS-A has two trusted sources for user accounts/user profiles:

- Member Data: Soldier data is populated to HCM via an external interface from HRC. The member's self-service account is provisioned within HCM. The self-service account is then published to ELM and CRM.
- POI Data: The POI data is bulk loaded in the initial load via PeopleSoft Component Interface (CI). After the initial load, POIs are manually added to HCM by HR Professionals.

6 MANAGING USER REQUEST PROCESS AND WORKFLOW APPROVAL

The user access request lets users request entities such as roles, resources, or entitlements. This functionality allows the ability to approve or deny these requests. Approval workflows will route to appropriate approvers (see step 3) and go to another level if there are segregation of duty (SOD) issues.

Request Process Components

The user request process is comprised of the following six components:

1. Request: An entity (or task) that a user creates in HCM to perform an action that requires approval before that action can be performed.
2. Request Types: The user access request has five types of requests:
   a. User Management: create user, modify user, etc.
   b. Role & Row Security Management: assign roles to a user and assigns row security to a user.
   c. Account Management: enable account, disable account, modify account, etc.
   d. Entitlement Management: instate entitlement or revoke entitlement
   e. Provisioning: provision application instance and access policy based application instance provisioning
3. Workflow Approval level:
   a. Supervisor Approvers: Validates the users need for the request and assigns to the S1 Pool upon approval or denies the request.
   b. S1 Pool Approvers: Selects the Validator(s) to review and approve the user request.
   c. Validator: Reviews and approves all user requests. When there is an SOD violation, assesses impact and corrections violations/conflicts, or rejects request.
   d. Validator Admin: Reviews and approves any user requests that have SOD mitigation and justification. When there is an SOD violation, assesses impact and corrections violations/conflicts, or rejects request.
   e. Key Entity Enabling (KEEB) Validator: Reviews, modifies, and approves access requests for Subcategories within the FSA, and Data Security Categories.
   f. Principle Validator: Reviews, modifies, and approves access requests for KEEB Validator, and Subcategories within the Functional Sys-Admin Category
   g. Maintenance System Admin (MSA) Validator: Reviews, modifies, and approves program-level access requests.
4. Notification: Information regarding submitted requests for approval are reflected on the Access Request dashboard after specifying desired search parameters.
5. ELM Verification Check: Each request will be through an ELM verification check process, which confirm the requestor has the required training certification for the requested role.
6. SOD Verification Check: Each request will go through a SOD verification check process, which confirms that there are NO conflicting roles granted to the requestor.

7 AUDIT MANAGEMENT

To implement a valuable Audit Management Plan, IPPS-A must develop and maintain a set of policies to support IT strategy. These policies should include policy intent; roles and responsibilities; an exception process; a compliance approach; and references to procedures, standards, and guidelines.

The scope of the IAM Auditing function includes auditing the following:

- The authentication process should include a risk assessment of the sensitivity of information available to users, locations from which the users may request access (for example, secure internal network or non-secure public network) and selecting an authentication process.
- A unique identifier is used to track the initiator of every transaction to investigate the initiator for legal or operational reasons. The initiator can be either a Soldier, POI, or contractor.
- The identity of the originating server should uniquely authenticate itself to the destination server, typically by presenting a digital certificate that is an unforgeable, electronic proof of identity.
- Access control policy defines an organization's access controls and establishes how often passwords must be changed; password complexity and history reduce the risk of a successful password "hacking" attack; and limitations on, or logging of, administrator activities with so-called super user privileges (system and database administrators) who may be able to bypass traditional controls.
- User provisioning transactions include approval workflows to create new users; to ensure that when users' job functions change, their corresponding access privileges are changed to align with their new job functions; and to ensure that terminated users' access privileges are removed immediately upon termination.
- Role-management transactions specify roles from initiation to revocation as user profiles are managed. Role profile audits cover changes to role profile attributes, role administrators, and other role groups.
- Mobility impact considers that users can connect to applications from mobile devices such as smart phones and tablet computers. Attributes associated with mobile devices may be different. All relevant information that can uniquely identify a mobile transaction is captured in the audit.

7.1 Segregation of Duty

The compliance assessment ensures that compliance rules and initiatives are maintained to ensure that no new risks are introduced into the environment. Implementation of SOD is based on verifying that no role conflicts will exist if requested roles are granted.

Access requests that a user submits are checked using predefined rules. If no conflicts, the request is not flagged as an SOD violation. An SOD violation occurs when a user requests access to a program level role (administrative), with an end user role (functional), which would cause a role violation. All conflicts must be mitigated by the appropriate validator prior to the final approval of the access request.

8 TESTING STRATEGY

Provisioning/Application Security testing is performed during Developmental and Operational testing to determine compliance with applicable IT security controls. Testing IPPS-A components complies with DoD and Army requirements that all systems be certified and accredited in accordance with Department of Defense Instruction (DoDI) 8510.01, Risk Management Framework (RMF), and CDRL 03-001, IPPS-A Application Security Plan. Additionally, the Program Management Office (PMO) coordinates with the SI Test Team, Operational Test Community, and Director Operational Test and Evaluation (DOT&E) to ensure that all appropriate external Provisioning/Application Security events are conducted in accordance with the DOT&E Cybersecurity Memorandum and Cybersecurity Assessment Handbook.

8.1 Test Script/Scenario Development Standard

The SI Testing Team develops test scripts to verify low-level system specifications; whereas test scenarios validate that the application meets the user's business process needs. Test scripts/ scenarios are developed using approved development standards and approved templates. The testing strategy consists of four phases for each release, and within each phase is a set of one or more test events. See Table 8-1.

Table 8-1 Test Phases

Test Phase	Events
Integration Testing	Unit-Level Testing (ULT) and Design and Development Testing events
System Developer Testing	Product-Level Testing (PLT) and Developer's Integration Test (DIT) events
Government Acceptance Testing (GAT)	Capability Acceptance Test (CAT), System Integration Test (SIT), Software Qualification Test (SQT), and System Acceptance Test (SAT) events
Operational Testing	Limited User Test (LUT) and Initial Operational Test and Evaluation

8.2 Provision Testing

Manual testing and a test utility are used to conduct provisioning tests. Testing utilizes PeopleSoft to provision a user to a target system to test basic provisioning operations (create, update, or deactivate) on configurable target resources.

Additional testing to perform includes:

- Test Basic Connectivity: Tests the connection to the target system using the IT resource for the target system.
- Test Provisioning: Performs a basic "create user" operation on the target system.
- Test Reconciliation: Performs a basic reconciliation operation on the target system.
- Test ELM: Determines/verifies ELM, the verification will confirm the requestor has the required training certification for the requested role.
- Test SOD: Determines/verifies whether assigned roles trigger SOD violations and the resolution logic.

9 DEPLOYMENT STRATEGY

Note: This section is pending update from Sustainment and Deployment team.

Appendix A: Acronyms and Abbreviations

The following acronyms and abbreviations may appear in this document:

Acronym	Definition
AACG	Application Access Controls Governor
AAES	Army Approved Enterprise Service
ABAC	Attribute-Based Access Control
AC	Active Component
ACC	Army Contracting Command
ACID	Atomicity, Consistency, Isolation, Durability
ACL	Access Control Lists
ACSIM	Assistant Chief of Staff for Installation Management
ADF	Application Development Framework
AES	Army Enterprise Service
AFCOS	Automated Fund Control Order System
AKO	Army Knowledge Online
ALM	Application Lifecycle Management
ALT	Alternate Login Token
ALTESS	Acquisition, Logistics and Technology Enterprise Systems and Services
AMRDEC	Aviation and Missile Research, Development and Engineering Center
AO	Authorizing Official
AOS	Army Organization Server
API	Application Program Interface
APR	Approval
AR	Army Regulation
ARISS	Army Recruiting Information Support System
ARNG	Army National Guard
ASD	Application Security Design
ASIP	Army Stationing and Installation Plan
ASORTS	Army Status of Readiness and Training System
ASP	IPPS-A Application Security Plan
ATCTS	Army Training and Certification Tracking System
ATMS	Army Training Management System
ATN	Army Training Network
ATRRS	Army Training Requirements and Resources System
AUP	Acceptable Use Policy
AV/DF	Audit Vault Database Firewall
AWOL	Absent Without Official Leave
BEN	Benefits
BI	Business Intelligence
BIDA	Business Intelligence and Data Analytics Plan
BIDE	Basic Identity Data Elements
CA	Computer Associate

Acronym	Definition
CAC	Common Access Card
CAT	Capability Acceptance Testing
CATS	Combined Arms Training Strategies
CCIMM	Cadet Command Information Management Module
CCM	Continuous Control Monitoring
CDRL	Contract Data Requirements List
CE	Computing Environment
CI	Component Interface
CL	Confidentiality Level
CLM	Contract Lifecycle Management
CM	Configuration Management
CMP	Configuration Management Plan
CNSSI	Committee on National Security Systems Instruction
COOP	Continuity of Operations Plan
COTS	Commercial Off-the-Shelf
CRM	Customer Relationship Management
CRUD	Create, Read, Update, and Delete
CSR	Change/Service Request
DAPMIS	Department of the Army Photograph Management Information System
DBA	Database Administrator
DBUM	Database User Management
DCIPS	Defense Casualty Information Processing System
DDL	Data Definition Language
DEERS	Defense Enrollment Eligibility Reporting System
DFAS	Defense Finance and Accounting Service
DISA	Defense Information Systems Agency
DIT	Developers Integration Testing
DJMS-RC	Defense Joint Military Pay System – Reserve Component
DMDC	Defense Manpower Data Center
DML (SQL)	Data Manipulation Language
DML (Army)	Distributed Manning Level
DMN	Domain
DMSL	Distribution Management Sub-Level
DOC	Document
DoD	Department of Defense
DoDAF	Department of Defense Architecture Framework
DoDD	Department of Defense Directive
DoDI	Department of Defense Instruction
DOT&E	Director, Operational Test and Evaluation
DRSS-A	Defense Readiness Reporting System – Army
DS Logon	Department of Defense Self-Service Logon
DTMS	Digital Training Management System

Acronym	Definition
EAMS-A	Enterprise Access Management Service - Army
EBS	E-Business Suite
ECR	Engineering Change Request
EDIPI	Electronic Data Interchange Personal Identifier
EE	Enterprise Edition
EJB	Enterprise Java Beans
EJS	Enterprise Job Scheduler
ELM	Enterprise Learning Management
EME	Enterprise Meta Environment
eMILPO	Electronic Military Personnel Office
ERB	Enlisted Record Brief
ERP	Enterprise Resource Planning
ESB	Enterprise Service Bus
ETCG	Enterprise Transaction Controls Governor
ETL	Extract Transform Load
EUCOM	US European Command
FG	Financial Governance
FIAR	Financial Improvement Audit Readiness
FIPS	Federal Information Processing Standard
FISCAM	Federal Information System Controls Audit Manual
FISMA	Federal Information Security Management Act
FMD	Functional Management Division
FOUO	For Official Use Only
FPS	French Public Sector
FSGLI	Family Service Members' Group Life Insurance
GAT	Government Acceptance Testing
GDE	Graphical Development Environment
GTC	Generic Technology Connector
GUI	Graphical User Interface
HCM	Human Capital Management
HIPAA	Health Insurance Portability and Accountability Act
HQDA	Headquarters, Department of the Army
HR	Human Resources
HRC	Human Resources Command
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IA	Information Assurance
IAM	Identity Access Manager
IAO	Information Assurance Officer
IB	Integration Broker
ICC	Integrated Circuit Chip

Acronym	Definition
ICF	Identity Connector Framework
ID	Identification
IDE	Integrated Development Environment
IDM	Identity Management
IDP	Identity Provider
IDT	Inactive Duty Training
IM	Information Management
IMS	Integrated Master Schedule
IP	Program Indicator
IPPS-A	Integrated Personnel and Pay System - Army
ISACA	Information Systems Audit and Control Association
ISSE	Information Systems Security Engineer
ISSM	Information System Security Manager
ISSO	Information System Security Officer
IT	Information Technology
ITE	Integrated Test Environment
J2EE	Java 2 Platform Enterprise Edition
JCS	Joint Chiefs of Staff
JNDI	Java Naming Directory Interface
KEEB	Key Entity Enabling
LDAP	Lightweight Directory Access Protocol
LUT	Limited User Testing
MAC	Mission Assurance Category
MCU	Multi-Component Unit
MDS	Metadata Services
MFA	Multifactor Authentication
MOBODEE	Mobilization, Operations, Deployment, Employment, and Execution
MOS	Military Occupational Specialty
MS	Microsoft Software
MSA	Maintenance System Admin
MyERB	My Enlisted Record Brief
NET	New Equipment Training
NIPRNET	Non-Secure Internet Protocol Router Network
NIST	National Institute of Standards and Technology
NSA	National Security Agency
OAM	Oracle Access Manager
OASIS	Organization for the Advancement of Structured Information Standards
OBIA	Oracle Business Intelligence Applications
OBIEE	Oracle Business Intelligence Enterprise Edition
OE	Organizational Element
OEM	Oracle Enterprise Manager
OHS	Oracle HTTP Server

Acronym	Definition
OIAM	Oracle Identity Access Management
OLTP	On-Line Transaction Processing
OOB	Out-Of-Band
OPAM	Oracle Privileged Account Manager
OPM	Office Personnel Management
OPRCLASS	Operator Class
ORB	Officer Record Brief
OSB	Oracle Service Bus
OSG	Operational Security Guide
OTJ	On-the-Job
OTSG	Office of the Surgeon General
OUD	Oracle Unified Directory
PAR	Personnel Action Request
PD	Provisioning Design
PDC	Primary Data Center
PDF	Portable Document Format
PDP	Policy Decision Component
PDR	Person Data Repository
PEA	Preventive Enforcement Agent
PEO EIS	Program Executive Office Enterprise Information System
PEP	Policy Enforcement Point
PIA	PeopleSoft Internet Architecture
PIH	PeopleSoft Interaction Hub
PII	Personally Identifiable Information
PIN	Personal Identification Number
PIP	Personnel Identity Protection
PIV	Personal Identity Verification
PKI	Public Key Infrastructure
PLT	Product Level Testing
PM	Program Manager
PMO	Program Management Office
PMP	Project Management Plan
POA&M	Plan of Action and Milestones
POC	Point of Contact
POI	Person Of Interest
POSC-Edit	Personnel Occupational Specialty Code-Edit
PP	Provisioning Plan
PPA	Personnel Processing Activity
PPIP	Program Protection Implementation Plan
PPP	Program Protection Plan
PRD	Production
PRR	Payroll Risk Reduction

Acronym	Definition
PS	PeopleSoft
PSC	Personnel Service Company
PY	Payroll
QA	Quality Assurance
RA	Regular Army
RAPIDS	Real-Time Automated Personnel Identification System
RBAC	Role Based Access Control
RC	Reserve Component
RCAS	Reserve Component Automation Systems
RCMS-G	Reserve Component Manpower System-Guard
RDBMS	Relational Database Management System
RMF	Risk Management Framework
ROTC	Reserve Officers' Training Corps
ROWSECCLASS	Data Level Security
RP	Relying Party
RPAC	Reserve Personnel Action Centers
RPG	Roles and Permissions Guide
S/SS	System/Subsystem Specifications
S/SSs	System/Sub-System Specifications
SAAR	System Authorization Access Request
SAML	Security Assertion Markup Language
SAML2	Security Assertion Markup Language Version 2
SAT	System Acceptance Test
SDC	Secondary Data Center
SDLC	System Development Lifecycle
SDT	System Development Testing
SFPA	Suspension of Favorable Personnel Actions
SFTP	Secure File Transfer Protocol
SGLI	Service Members' Group Life Insurance
SI	System Integrator
SIPRNet	Secret Internet Protocol Router Network
SIT	System Integration Test
SJT	Security Join Table
SME	Subject Matter Experts
SOA	Service-Oriented Architecture
SOAP	Service-Oriented Architecture Protocol
SOAPUI	Simple Object Access Protocol User Interface
SOD	Segregation of Duties
SOP	Standard Operating Procedure
SOX	Sarbanes-Oxley Act
SP	Sensitive Products
SPML	Service Provisioning Markup Language

Acronym	Definition
SPRs	System Problem Reports
SQL	Structured Query Language
SQLID	Structured Query Language Identification
SQR	Structured Query Report
SQT	Software Qualification Test
SRB	Soldier Record Brief
SS	Self-Service
SSL	Secure Sockets Layer
SSLI	State Sponsored Life Insurance
SSN	Social Security Number
SSO	Single Sign-On
SSP	System Security Plan
STB	Stonebranch
STIG	Security Technical Implementation Guide
SV	System View
SYNC	Synchronization
TAPD-AE	Total Army Personnel Database – Active Enlisted
TAPD-R	Total Army Personnel Database – Reserve
TEMP	Test and Evaluation Master Plan
TESS	Table Edit Support System
TIRS	Test Incident Report
TOPMIS	Total Officer Personnel Management Information System
UAT	User Acceptance Testing
UI	User Interface
UIC	Unit Identification Code
ULT	Unit-Level Testing
UMS	User Messaging Service
UPK	User Productivity Kit
URL	Uniform Resource Locator
US	United States
USACC	U.S. Army Cadet Command
USAR	United States Army Reserve
User ID	User Identification
VDI	Virtual Desktop Infrastructure
WF	Workflow
WIPT	Working Integrated Project Team
WLS	WebLogic Server
WLST	WebLogic Scripting Tool
WML	Wireless Markup Language
WS-SECURITY	Web Service Security
XML	Extensible Markup Language

Appendix B: CACI Consolidated Reference Documents

This appendix lists other referenced documents that complement required CDRLs.

Document #	PWS Reference	Title
03-001	4.3	IPPS-A Application Security Plan
03-004	4.3	IPPS-A Application Security Design
03-006	4.3.1	IPPS-A Provisioning Design
03-007	4.3.2	IPPS-A Roles and Permissions Guide
14-001	4.13.5.2	Deployment Plan

Appendix C: Glossary

The following terms clarify content in this document:

Term	Definition
Access Control	The process of granting or denying specific request to: 1) obtain and use information and related information processing services; and 2) enter specific physical facilities (e.g., federal buildings, military establishments, and border crossing entrances). SOURCE: FIPS 201; CNSSI-4009
Advance Security	Application of advanced security solutions Oracle Audit Vault, Database Vault and Label Security
Agent Types	Types of adversarial operative that counterintelligence and cybersecurity members define as threats.
Anarchist	Someone who rejects all forms of structure, private or public, and acts with few constraints
Application	All COTS products that make up the functional IPPS-A system.
Attribute-Based Access Control (ABAC)	An access control paradigm whereby access rights are granted to users by using policies that combine attributes. Policies can use any type of attribute (user, resource, environment, etc.)
Auditing	Formal internal and external inspections and verifications to check whether DoD standards, guidelines, and policies are being followed, data capture and reporting is accurate, and all key performance indicator targets are being met.
Authentication	Authentication is the process of determining that a user's claimed identity is legitimate. This usually involves a user ID and password, but can include any other method of demonstrating identity, such as a smart card, retina scan, voice recognition, or fingerprints.
Authentication factor	A category of credential for identity verification. For Multi-Factor Authentication, each additional factor increases the assurance that an entity involved in some communication or requesting access to a system is who or what they are declared to be. The three most common categories are something you know (knowledge), something you have (possession) and something you are (inherence).
Authorized User	User with appropriate roles and permissions in excess of self-service.
Availability	Ensuring timely and reliable access to and use of information.
Business Rules	A rule that defines or constrains some aspect of a business structure.
Category	The program vision is comprised of three end user categories: Member, Commander/Manager, and HR Professionals. All Soldiers will be Members and can only access their record and information. Commanders/ Managers will have access to their subordinate records and have the ability to approve/transact. HR Professionals will have expanded access to act as HR business partners at various echelons.
Centric	Having (something) as a central interest, influence, subject, etc.
Civil Activist	Highly motivated but non-violent supporter of cause
Competitor	Business adversary who competes for revenues or resources (acquisitions, etc.)
Component Interface (CI)	A PeopleSoft CI is a PeopleTools definition created in PeopleSoft Application Designer. It enables synchronous access to a PeopleSoft component from another application. CIs can integrate PeopleSoft with another PeopleSoft application or external systems.
Confidentiality	Preserving authorized restrictions on access and disclosure, including means for protecting privacy and proprietary information.
Corrupt Government Official	Person who inappropriately uses his or her position within the government to acquire company resources
Credential	An object that authoritatively binds an identity (and optionally, additional attributes) to a token possessed and controlled by a person.
Cyber Vandal	Derives thrills from intrusion or destruction of property, without strong agenda
Data Masking	A method of creating a structurally similar but inauthentic version of an organization's data for purposes such as software testing and user training to protect actual data with a functional substitute when real data is not required
Data Miner	Professional data gatherer external to the company (includes cyber methods)

Term	Definition
Data Permission	Data Permissions are derived from the combination of Security Sets and Access Types in order to secure different types of data element based on selected criteria.
Data Permission Security	Controls the application data that a user is allowed to access in the PeopleSoft system. With application data security, data permissions can be set at the table level, row level, and field level.
Data Redaction	Suppressing sensitive data, such as any personally identifiable information (PII) that can be used on its own or with other information to identify or locate a single person or identify an individual in context
Defense in Depth	The practice of layering defenses to provide added protection to the component or software by placing multiple barriers between an attacker and the IPPS-A information resources. Defense in depth increases security by raising the effort needed in an attack to be successful.
Digital Identity	A set of identifiers (user attributes, characteristics) by which an entity is recognizable and distinguishable from any other entity.
Digital Signature	An asymmetric key operation in which the private key is used to digitally sign data and the public key is used to verify the signature. Digital signatures provide authenticity protection, integrity protection, and nonrepudiation. (Source: SP 800-63)
DoD Self-Service Logon (DS Logon)	A secure, self-service logon ID created by the Defense Manpower Data Center (DMDC) as an enterprise identity credential that grants individuals affiliated with the DoD or Department of Veterans Affairs (VA) access to websites using a single user ID and password. DS Logon supports the Personnel Identity Protection (PIP) Directive and National Institute of Standards and Technology (NIST) e-Authentication guidance providing high-level authentication assurance where CAC authentication is not available.
Elevated Access Request	Process within IPPS-A functionality where users can request elevated system privileges.
Employee, Disgruntled	Current or former employee with intent to harm the company
Entity	A subject (active element that operates on information or the system state) or an object (passive element that contains or receives information).
External User	A provisioned IPPS-A user accessing the system via a non .mil network
Federated Authentication	Allows members of one organization to use their authentication credentials to access a web application in another institution
Foundation Tables	Key tables to implement an enterprise-wide HRMS, which provide a structure to establish and maintain core organizational data, eliminate redundant data entry by providing default values, and help enforce data integrity by ensuring consistency. Foundation tables must be established for capturing data or processing personnel and payroll transactions in the system
GoldenGate	An Oracle application that allows replication, filtering, and transformation of data from one database to another database.
Governance	Ensures that DoD/Army needs, conditions, and options are evaluated to determine balanced, agreed-upon enterprise objectives to be achieved; setting direction through prioritization and decision-making and monitoring performance and compliance against agreed-upon direction and objectives.
Government Cyber Warrior	State-sponsored attacker with significant resources to affect major disruption on national scale
Government Spy	State-sponsored spy as a trusted insider, supporting idealistic goals
Groups	A collection of users to permit access to common functionality
HR Professional	A subset of authorized users working in Human Resources.
Identity Federation	An arrangement that can be made among multiple enterprises that lets subscribers use the same identification data to obtain access to the networks of all enterprises in the group
Information System Security Officer (ISSO)	Individual assigned responsibility for maintaining the appropriate operation security posture for an information system or program.
Integrity	Guarding against improper information modification or destruction and includes ensuring information non-repudiation and authenticity by always passing the ACID Test (Atomicity, Consistency, Isolation, Durability).
Internal Spy	Professional data gatherer as a trusted insider, generally with a simple profit motive
Internal User	A provisioned IPPS-A user accessing the system via the .mil network

Term	Definition
Irrational Individual	Someone with illogical purpose and irrational behavior
Java Naming & Directory Interfaces (JNDI)	JNDI is a service that facilitates resources on a server to be referenced by a generic name by various applications, system processes, etc.
Keystore	A database that holds encrypted information and information on how to decrypt it.
Least Privilege	Limiting access through system administration to only authorization required to perform required functions. The goal is to reduce risk by limiting the number of people with access to critical system security controls—controlling who is allowed to enable or disable system security features or change privileges of users or programs.
Legal Adversary	Adversary in legal proceedings against the company, warranted or not
Member	A Soldier.
Metadata	Data that describes other data.
Mobster	Manager of organized crime organization with significant resources
Need-to-Know	The practice limiting authorized access to sensitive information where individuals whose need for the information has been verified.
Nonrepudiation	Assurance that the sender of information is provided with proof of delivery and the recipient is provided with proof of the sender's identity, so neither can later deny having processed the information. (Source: SP 800-60)
Notification	A method or means of delivering a message to a set of recipients. Notification system complexity often depends on organizational requirements and/or the types of messages sent. Types of notification options are *email* and *worklist* notifications (list of task assigned to a user).
Object	Passive information system-related entity (e.g., devices, files, records, tables, processes, programs, domains) containing or receiving information. Access to an object (by a subject) implies access to the information it contains.
Operation-Level Workflow	Workflow associated with operations that OUD is to grant to a user through a request.
PeopleCode	A proprietary language used by the PeopleSoft Enterprise component processor; it generates results based on existing data or user actions. Using various PeopleTools tools, external services are available to all PeopleSoft Enterprise applications wherever PeopleCode can be executed.
PeopleSoft User Synchronization	User profile synchronization involves setting up each PeopleSoft database in the enterprise to send and receive user profile updates through the Integration Broker. When you enter new profiles or modify and delete existing profiles on any publishing database and save, PeopleCode publishes a user profile service operation—which contains a user profile message—and routes the message to all subscribing nodes according to your specifications. The subscribing databases then update the user profile data with data from the publishing database.
Permission Lists	Groups of authorizations that are assigned to roles. Permission lists store sign-in times, page access, PeopleTools access, and so on. A permission list may contain one or more types of authorizations. The fewer types of permissions in a permission list, the more modular and scalable the implementation.
Person of Interest (POI)	An authorized user in IPPS-A who is not an employee (Member).
Personally Identifiable Information (PII)	Any information about an individual (such as name, social security number, biometric records) that can be used to recognize or trace an individual's identity alone, or when combined with any other personal information linked or linkable to an individual
Privileged User	A user who is authorized (and therefore trusted) to perform security functions or who has access to system control, monitoring, and administration functions (system administrator, database administrator, application administrator, security administrator, Information System Security Officer [ISSO], system programmers/developers, etc.)
Process Group	A group of application processes (performed in a defined order) that users can initiate in real time, directly from a transaction entry page.
Process Scheduler	Enables the scheduling of recurring processes, the creation of jobs, scheduling and submitting process requests, managing batch processes, and reports.

Term	Definition
Provisioning	Providing users with access to data and technology resources with appropriate roles and security attributes to perform assigned duties.
Radical Activist	Highly motivated, potentially destructive supporter of cause
Repository	In software development, a central file storage location used by version control systems to store multiple versions of files.
Request-Level Workflow	Workflows initiated by request service in SOA and performed as a part of a parent request. Contains High-Level information associated with the entire request.
Resource Type	Different application URIs to that a user can access.
Risk Management Framework (RMF)	An integrated organization-wide program for managing information security risk to organizational operations (i.e., mission, functions, image, and reputation), organizational assets, individuals, other organizations, and the Nation, resulting from the operation and use of federal information systems. The RMF replaces the DoD Information Assurance (IA) Certification and Accreditation Process (DIACAP) and manages the lifecycle cybersecurity risk to DoD information systems.
Role-Based Access-Control (RBAC)	A policy-neutral access control mechanism defined around roles and privileges
Roles	Intermediate objects that link user profiles to permission lists. The system can assign multiple roles to a user profile and can assign multiple permission lists to a role.
Row-Level Security	Controls access to individual rows of data stored within database tables through security views; enables you to specify the rows of data that a particular user is permitted to access.
Security Access Type	Security access types are ways of securing the data within a security set. Each security set has a number of security access types that can be enabled
Security Assertion Markup Language (SAML)	Security Assertion Markup Language is an XML-based, open-standard data format for exchanging authentication and authorization data between parties, in particular, between an identity provider and a service provider. SAML is a product of the Organization for the Advancement of Structured Information Standards (OASIS) Security Services Technical Committee.
Security Set	Security sets are groupings of data that is being secured. The sets differ by the origin of the transaction security data.
Segregation of Duties	A basic internal control that prevents or detects errors and irregularities by assigning to separate individual responsibility for initiating and recording transactions and custody of assets. Commonly used in large IT organizations so that no single person is in a position to introduce fraudulent or malicious code without detection.
Sensationalist	Attention-grabber who may employ any method for notoriety; looking for “15 minutes of fame”
SetID	An identification code that represents a set of control table information or TableSets
Single Sign-On (SSO)	The process that allows a user to access multiple related but independent software systems or applications with one set of login credentials. The process validates the user for all applications to which they have access and eliminates further prompts when they switch software systems or applications during a single session.
Stale Request	A request that has failed, most likely because of a failed scheduled task (either because the Quarts trigger is failing/still processing, or the previous jobs failed).
Subcategory	Subcategories are groupings of capabilities within IPPS-A for a user to conduct his/her duties.
Subject	Generally an individual, process, or device causing information to flow among objects or change to the system state.
System Development Lifecycle (SDLC)	Scope of activities associated with a system, encompassing system initiation, development and acquisition, implementation, operation and maintenance, and ultimate disposal that instigates another system initiation. Source: SP 800-34; CNSSI-4009
TableSet	A means of sharing similar sets of values in control tables, where actual data values are different but the table structure is the same
TableSet Sharing	Shared data stored in many tables based on the same TableSets. Tables that use TableSet sharing contain the SETID field as an additional key or unique identifier
Target System	The destination system/resource where user and role are provisioned.
Terrorist	Person who relies on the use of violence to support personal socio-political agenda
Thief	Opportunistic individual with simple profit motive

Term	Definition
Time-out	A fixed period of time before a system disconnects a session, which occurs when the system detects no user activity
Transaction	An event initiated by a user or system process that is committed as a data record in the database
Trusted Reconciliation	The name given to the original source of user, roles and role Membership
Trusted Source	A system with authoritative data that other systems can safely use.
Universal Resource Identifier	Universal Resource Identifier contains information about the component's location on the server.
User Account/User Profile	A visual display of attributes associated with a user that defines access authorizations.
User Profile / User Account	A visual display of attributes associated to a specific user that defines access authorizations.
Vendor	Business partner who seeks inside information for financial advantage over competitors
Virtual Desktop Infrastructure (VDI)	Virtualization technology that hosts a desktop operating system on a centralized server in a data center. A variation on the client-server computing model, sometimes referred to as *server-based computing*. Term was coined by VMware.
Workflow	Automated vital business tasks, integrating activities of multiple users into a single business process. Ties individual steps together so the system can coordinate activities
Worklist	An organized list of the work items awaiting a user's attention. Work items are selected from a worklist, and selecting an item automatically brings up the pages needed in order to work on it; bypassing any other navigational structure and work directly from the worklist.